

ESCNet: Edge-Semantics Collaborative Network With Multiscale Fusion for Land Cover Classification

Guangqi Xie , Jiawei Hu, Rui Tao, Zhiqi Zhang , *Member, IEEE*, and Shenggu Yuan 

Abstract—Land cover classification is a critical task in remote sensing image processing, with its core objective being the accurate classification of surface coverage through pixel-level feature extraction, thereby supporting applications such as environmental monitoring and land use planning. However, the inherent tension between the continuous physical nature of land cover and the discrete classification framework introduces semantic ambiguity at category boundaries and impedes effective model learning across scales. While edge features are commonly incorporated to mitigate boundary ambiguity, existing methods often ignore achieving synergistic integration between edge guidance and multiscale feature fusion, leading to suboptimal performance. To address these challenges, we propose an edge-semantics collaborative network (ESCNet). We design a dual-branch architecture to decouple feature extraction across scales, and introduce an edge-guided fusion mechanism that strategically leverages edge cues to direct the integration of features from the dual branches, thereby establishing synergistic collaboration between boundary localization and semantic representation. First, we design a direction-sensitive shuffle detail branch to enhance spatial detail representation capabilities through multidirectional convolutions and channel rearrangement. Concurrently, a spatial-channel dominated semantic branch is constructed to explore synergistic spatial-channel contextual information using linear attention mechanisms. Additionally, an edge-guided feature fusion decoder is introduced to reduce fusion deviations between semantic and detail features by leveraging edge features as constraints. Experimental results on the DeepGlobe and LoveDA datasets demonstrate that ESCNet achieves mIoU scores of 73.9% and 59.32%, respectively, showing certain advantages in the preservation of fine-grained structures and semantic consistency.

Index Terms—Dual-branch network, edge guidance, land cover classification, semantic segmentation.

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I. INTRODUCTION

HUMAN activities and environmental alterations have resulted in significant impacts on the Earth's surface composition [1]. Efficient and accurate land cover classification algorithms are essential for promptly obtaining land cover information to meet the needs of decision-makers and landscape planners. With the continuous advancement of Earth observation technologies, a global constellation of Earth observation satellites with wide coverage and short revisit periods has been established. This enables us to rapidly acquire large amounts of high spatial resolution, temporal, and spectral resolution remote sensing images [2].

Land cover classification, formulated as a dense pixel-wise semantic segmentation task, requires assigning categorical labels to individual pixels within remote sensing imagery. This classification process inherently captures both biophysical characteristics (e.g., vegetation types, hydrological features) and anthropogenic patterns (e.g., urban structures, agricultural layouts) of land surfaces. The field of semantic segmentation has evolved significantly since the introduction of fully convolutional networks [3]. Convolutional neural networks (CNNs) gradually replaced conventional segmentation techniques [4], [5], owing to their superior hierarchical feature learning capacity and scalability on large-scale geospatial datasets. Subsequent architectural innovations, including encoder-decoder structures [6], attention mechanisms [7], and transformer-based frameworks [8], have demonstrated strong generalizability across diverse applications. Similarly, specialized approaches for land cover mapping have emerged [9], [10], focusing on spatial-contextual modeling through graph-based correlation or multiscale feature fusion. With the continuous improvement of image resolution of land cover classification, while enhancing the ability of detail representation of ground objects, it also exposes the inherent contradiction between local fine discrimination and global context integration of existing models due to the coexistence of misclassification of granular objects and semantic hollow in wide-area homogeneous areas. Therefore, there are still some ongoing challenges in the segmentation of land cover classification datasets.

The first challenge lies in handling vastly different object sizes in geospatial data, which hinders consistent multiscale analysis in land cover classification. Although the standard convolution focusing on local details method can capture the texture features of small and micro surface objects (e.g., isolated vegetation patches, broken wetlands), it is easy to produce irregular

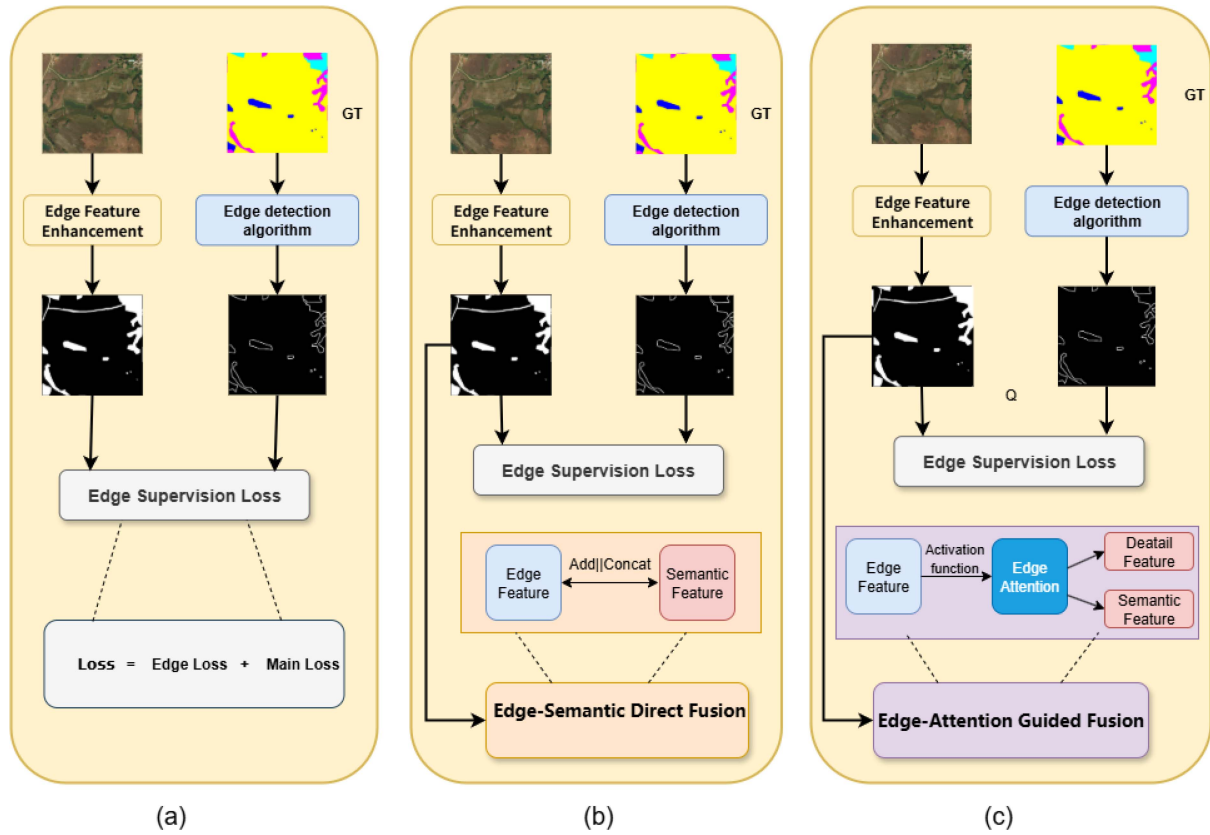


Fig. 1. Motivation of the proposed method. (a) Treating the edge loss as a component of the overall loss to enhance edge localization through backpropagation. (b) Recent methods further fuse edge features with semantic features via addition or concatenation. (c) Proposed method utilizes edge features solely for guidance rather than allowing them to directly participate in fusion.

segmentation results at the boundary of continuous geographical elements, resulting in a decline in the segmentation accuracy of land cover products. On the contrary, for large-scale continuous ground objects (e.g., water bodies), some deep learning (DL) methods use dilated convolution to expand receptive fields to improve the ability of global feature extraction [11] [12], but this strategy will weaken the model's fine recognition ability for small and microground objects. Building upon DeepLabv3+'s seminal atrous spatial pyramid pooling (ASPP) [13] module for multiscale context aggregation, later innovations [14] have largely addressed the cross-scale fusion challenge through novel architectural mechanisms. MSDF-net [15] uses multiscale large kernel convolution blocks to combine large kernel convolution with heterogeneous operations of partitioning, enhancing multiscale feature representation. OSHFNet [16] dynamically adjusts the offset direction of the convolution kernel, enabling it to adaptively deform along the path of the tubular structure and cover a longer continuous area. And the adoption of self-attention mechanisms [17] offers improved long-range dependency modeling. Segmenter [18] leverages the global self-attention mechanism of Vision Transformers (ViTs) to capture comprehensive contextual relationships across entire images. Although this approach achieves superior performance in modeling holistic scene understanding, its computational complexity scales quadratically with input resolution, posing substantial challenges for high-resolution geospatial imagery processing. SegFormer [19] addresses this efficiency limitation

through a hierarchical Transformer architecture combined with a lightweight multilayer perceptron decoder. This design reduces self-attention complexity to a linear scale via strategic feature sequence reorganization. However, its reliance on localized window-based attention mechanisms compromises cross-window semantic continuity, particularly evident in elongated geographical features such as river networks, where segmentation masks may exhibit fragmentation artifacts. In previous practices [20], [21], [22], many optimization methods have been designed to reduce the complexity of the self-attention mechanism, which can decrease the computational cost and improve inference speed without significantly compromising, or with only a slight reduction in, network performance. However, these approaches still struggle to achieve effective cross-scale contextual integration—particularly in harmonizing high-frequency boundary details with low-frequency semantic homogeneity.

The second challenge addresses the collaborative modeling dilemma between edge-aware methods and semantic segmentation. As illustrated in Fig. 1(a), some edge-aware approaches excel at precise edge localization but primarily treat detected edges as auxiliary loss terms within the global loss function, rather than unifying them with semantic reasoning through systematic integration [23]. Recent methods, as illustrated in Fig. 1(b), have increasingly emphasized synergistic collaboration through direct integration of edge features with semantic representations, where edge features actively participate in feature fusion via operations such as element-wise addition, channel

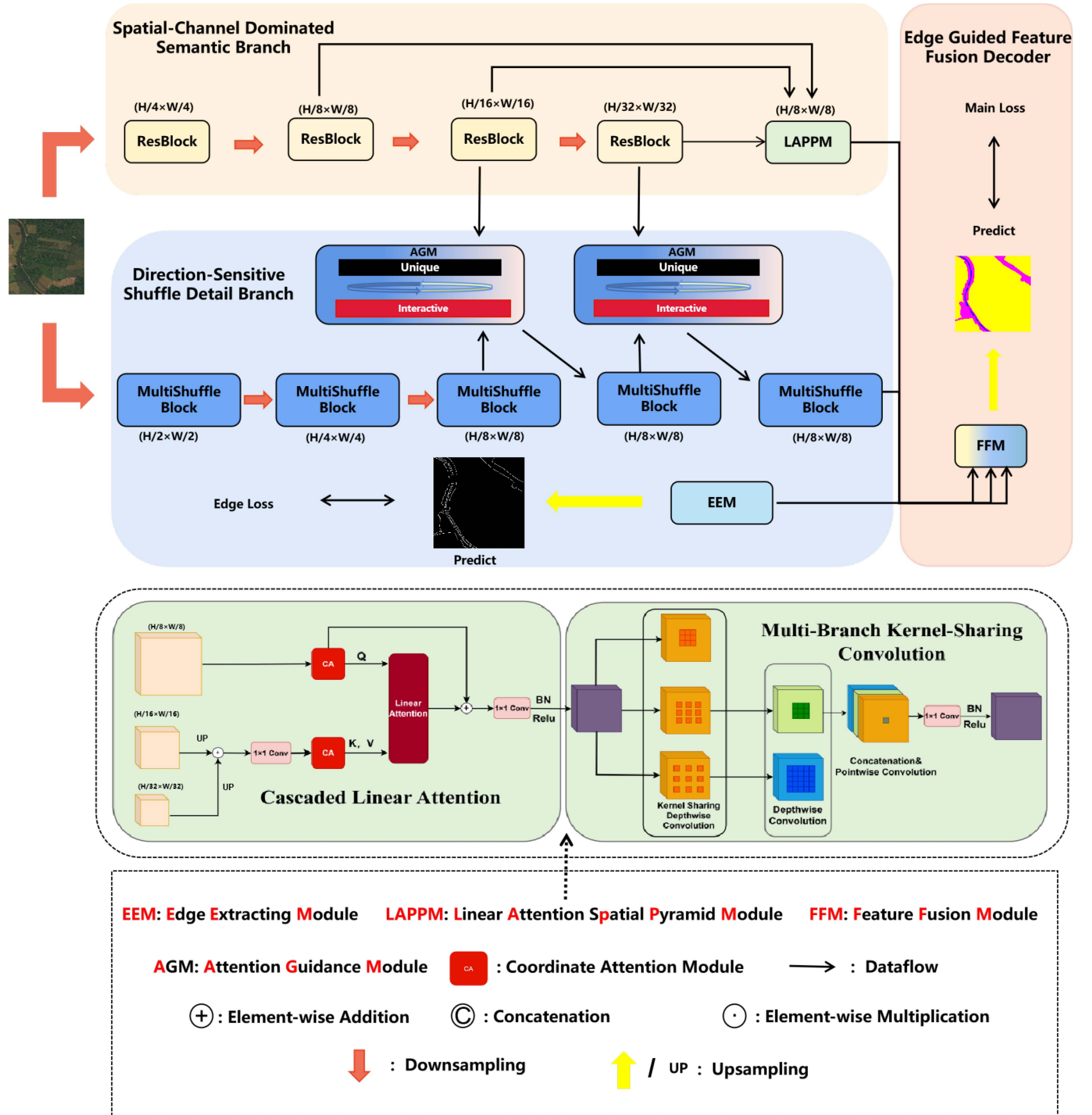


Fig. 2. Overall framework of ESCNet. The details of LAPPM are on the bottom.

concatenation, or other combination mechanisms, effectively leveraging their positional cues to enhance segmentation performance [24]. However, this direct fusion strategy exhibits inherent limitations: while edge-aware algorithms achieve enhanced accuracy in terrestrial boundary identification, the refined edge features lack discriminative guidance for classifying interior pixels of objects. This oversight frequently leads to misclassification of categories with well-defined boundaries but internally homogeneous surfaces, as edge information provides

no semantic cues for interior regions. To resolve this, as shown in Fig. 1(c), we transform the edge features into edge attention weights, thereby establishing an indirect guidance mechanism to regulate feature fusion while preserving structural-semantic consistency.

To address these two challenges, we propose a novel edge-semantics collaborative network (ESCNet). Specifically, ESCNet includes the direction-sensitive shuffle detail branch (DSSDB), which enhances details by introducing Asymmetric

convolution, and advanced semantics from another branch. The spatial-channel dominated semantic branch (SCDSB) strengthens the interaction between space and channels, acquiring multiscale information to address the issue of class imbalance in land cover classification datasets. Finally, we designed the edge-guided feature fusion decoder (EGFFD), using the acquired edge information to guide the fusion of the two branches.

The main contributions can be summarized as follows.

- 1) We propose ESCNet: a novel end-to-end dual-branch network to resolve the persistent challenge of blurred boundaries in land cover segmentation—stemming from the continuous physical nature of land surfaces and multiscale learning bottlenecks. This architecture extracts multiscale semantic cues and multidirectional spatial details in parallel, while additionally deriving dedicated edge features.
- 2) Due to the complex geospatial characteristics of land cover classification, we designed the DSSDB to capture multidirectional spatial details and edge information. Furthermore, to handle the diverse scale variations encountered in this task, we designed the SCDSB. It captures contextual information by leveraging interactions between spatial and channel dimensions.
- 3) EGFFD was proposed to fuse two branches by edge information. Extensive comparative experiments were conducted on two typical land cover classification datasets, validating that the proposed ESCNet significantly outperforms other mainstream semantic segmentation methods in land cover classification tasks.

The rest of this article is structured as follows. Section II reviews existing dual-branch segmentation methods and contextual feature extraction techniques. Section III introduces the proposed ESCNet architecture, detailing its core modules. Sections IV and V provide a comprehensive experimental analysis. Section VI discusses methodological insights, comparative advantages, and limitations. Section VII concludes the work and outlines future research directions.

II. RELATED WORK

A. Dual-Branch Segmentation Networks

In conventional encoder-decoder architectures, input features undergo progressive downsampling through convolutional layers to capture high-level semantic representations, followed by upsampling operations to reconstruct spatial resolution for pixel-wise prediction. However, this resolution reduction process inherently compromises the preservation of fine-grained spatial details necessary for precise segmentation. To address this limitation, contemporary research has adopted dual-branch architectures that concurrently process multiscale features through parallel pathways.

BiSeNet [25] introduced a dual-resolution framework comprising two pathways: the Context Path employs a deep backbone network (e.g., ResNet) for hierarchical semantic abstraction through successive downsampling, while the Spatial Path maintains shallow convolutional layers to preserve high-resolution spatial details. These complementary feature streams are adaptively fused through a specifically designed

feature fusion module (FFM), achieving an optimal balance between computational efficiency and segmentation accuracy. This architectural innovation has inspired subsequent advancements: ABCNet [26] introduces linear attention mechanisms to replace conventional dot-product attention, reducing computational complexity from $O(n^2)$ to $O(n)$ while maintaining contextual modeling capacity. EIGNet [27] explicitly incorporates boundary constraints during multiresolution integration. From a control theory perspective, PIDNet [28] implements proportional-integral-derivative controllers to dynamically regulate feature fusion processes, effectively preventing edge over-extension and improving boundary smoothness in land cover segmentation. Addressing cross-modal discrepancies, FrFENet [29] introduces weighted fractional Fourier enhanced fusion, and ABMDRNet [30] develops bidirectional difference minimization operators to align heterogeneous feature representations, which is further enhanced in MDRNet+ [31] through prefusion modality calibration techniques.

In general, whether for single-modal tasks or multimodal tasks, most methods typically calculate the weight relationship between two branches or design special loss functions. These approaches focus on information exchange between dual branches, overlooking the incorporation of third-party information as guidance. Our method introduces edge information to reduce the differences between information at different levels, aiming to achieve results with good interpretability and generalization ability.

B. Context Information Extraction

Semantic segmentation addresses the fundamental task of assigning categorical labels to individual pixels within an image. Given the inherent spatial dependencies among adjacent pixels, effective contextual modeling is critical for discerning inter-object relationships and refining foreground-background differentiation. Contemporary methodologies for contextual integration primarily focus on three complementary strategies: spatial context enhancement, channel-wise feature recalibration, and their synergistic combination.

Spatial context modeling emphasizes expanding receptive field to capture broader contextual patterns. While enlarging convolutional kernels could theoretically achieve this, practical computational constraints often render such direct approaches infeasible. Dilated convolutions address this limitation by strategically introducing spacing intervals within convolutional kernels, enabling exponential receptive field expansion with minimal parameter overhead. Encoder-decoder architectures [5] further enhance spatial context extraction by integrating multi-scale features through skip connections, harmonizing high-resolution shallow features with semantically rich deep representations. To address scale variability in remote sensing imagery, pyramid-based frameworks have become prevalent. The foundational spatial pyramid pooling (SPP) has evolved into specialized variants: PSPNet [32] employs a pyramid pooling module (PPM) with four progressively sized convolutional kernels, while DeepLabv2 [33] utilizes an ASPP module featuring parallel dilated convolutions with dilation rates of 6, 12, and

18. KDPNet [34] optimizes ASPP through depth-wise separable convolutions, achieving enhanced parameter efficiency. Recent innovations continue to demonstrate the efficacy of SPP-derived architectures [35], [36], where advanced implementations consistently achieve advanced performance in land cover segmentation tasks.

Complementary to spatial approaches, channel-wise contextual modeling dynamically recalibrates feature importance across channels. RSANet [37] uses recursive self-attention modules to iteratively refine global channel dependencies, while SENet [38] introduces a squeeze-and-excitation mechanism that adaptively weights channels through global spatial aggregation and fully connected layers. CANet [39] extends this paradigm by incorporating directional positional priors via polar coordinate transformations, yielding a 2.3% mIoU improvement over conventional channel attention methods in urban scene segmentation.

The hybrid architectures synergize spatial and channel contexts to achieve comprehensive feature representation. CBAM [40] implements parallel channel and spatial attention modules to jointly emphasize informative features across both dimensions. CFNet [41] adopts a dual-branch structure for simultaneous spatial-channel context extraction, whereas FrHSPL [42] deeply integrates fractional-order domain information with spatial and spectral information. While these integrated approaches outperform single-dimension methods, they often incur significant computational overhead and parameter inflation. To mitigate this tradeoff, our work introduces optimized multidimensional context modeling through linear attention mechanisms [26] and shared-kernel convolutions, effectively balancing segmentation accuracy with computational efficiency.

III. METHOD

This section presents the proposed network architecture in five parts. Section A first outlines the overall structure of ESCNet (as shown in Fig. 2). Section III-B then introduces the DSSDB module. Furthermore, Section III-C elaborates on the SCDSB module in detail. Section III-D focuses on the EGFFD, which is designed for feature fusion of the aforementioned modules. Finally, Section III-E describes our composite loss function.

A. Overall Framework

ESCNet's architecture is driven by the need to preserve fine-grained spatial details while capturing hierarchical semantic context—A tradeoff inherent to remote sensing segmentation tasks. The DSSDB focuses on retaining directional textures and object boundaries through asymmetric convolutions and channel shuffling, while simultaneously generating edge supervision signals via its edge extraction module (EEM). Parallel to this, the SCDSB progressively compresses resolution through Res-Blocks. Its core component, the linear attention pyramid pooling module (LAPPM), aggregates and fuses multiresolution features extracted from this branch. The EGFFD dynamically integrates multiscale features from both branches, leveraging edge priors to mitigate fusion artifacts and boundary ambiguities. The decoder employs edge-guided attention mechanisms to align semantic

and detail features, followed by linear attention for long-range dependency modeling, culminating in a segmentation head that restores predictions to the original input resolution.

B. Direction-Sensitive Shuffle Detail Branch

Spatial details encompass the intricate structural patterns, textural variations, and discriminative features inherent in high-resolution imagery, where increased pixel density inherently preserves finer granularity of visual information. While conventional approaches such as skip connections and zero-padding convolutions partially mitigate detail degradation in deep networks, they prove inadequate for remote sensing applications characterized by extreme scale diversity and anisotropic object distributions. To address these limitations, we propose the Multi-Shuffle block (see Fig. 3) as the cornerstone of the DSSDB. This module synergistically integrates horizontal (3×1) and vertical (1×3) asymmetric convolutions to explicitly model orientation-sensitive spatial patterns—critical for capturing linear infrastructures (e.g., road, railway, and canal) and geometric boundaries (e.g., agricultural field parcels and building footprints) prevalent in remote sensing imagery. Complementing this, channel shuffling operations [43] disrupt fixed channel correlations through systematic permutation, fostering cross-dimensional information integration while enhancing feature generalizability. The resultant representations demonstrate superior noise robustness and spatial selectivity, effectively amplifying discriminative details while suppressing irrelevant high-frequency artifacts.

Existing research [44] substantiates that hierarchical semantic representations derived from deep layers can resolve ambiguities and compensate for information loss in shallow detail features through contextual semantics. Building on this premise, we design an attention guidance module (AGM) (see Fig. 3) to enable dynamic knowledge transfer from the semantic to the detail.

The mechanism operates through two sequential stages.

- 1) *Cross-Branch Feature Alignment*: Pixel-wise similarity maps are computed via Hadamard product between semantic feature (SF) and detail feature (DF), capturing latent semantic correlations across resolution levels.
- 2) *Attention Weight Generation*: The similarity maps undergo nonlinear activation through sigmoid normalization, producing adaptive weights (σ) that govern feature fusion.

This process can be represented as

$$\sigma = \text{sigmoid}(\text{Conv}_{1 \times 1}(\text{DF}) \cdot \text{Conv}_{1 \times 1}(\text{SF})) \quad (1)$$

where " $\text{Conv}_{a \times a}$ " denotes the use of a convolutional kernel of size $a \times a$.

The AGM's attention weights (σ) enable dynamic feature recalibration based on spatial-semantic confidence levels. Elevated σ values indicate regions where high-level SFs dominate, leveraging their abstract representations to enforce global consistency and suppress local ambiguities. Conversely, diminished σ values prioritize detail features to preserve high-frequency textures and precise boundary delineations. This adaptive mechanism ensures contextually informed detail enhancement—SFs

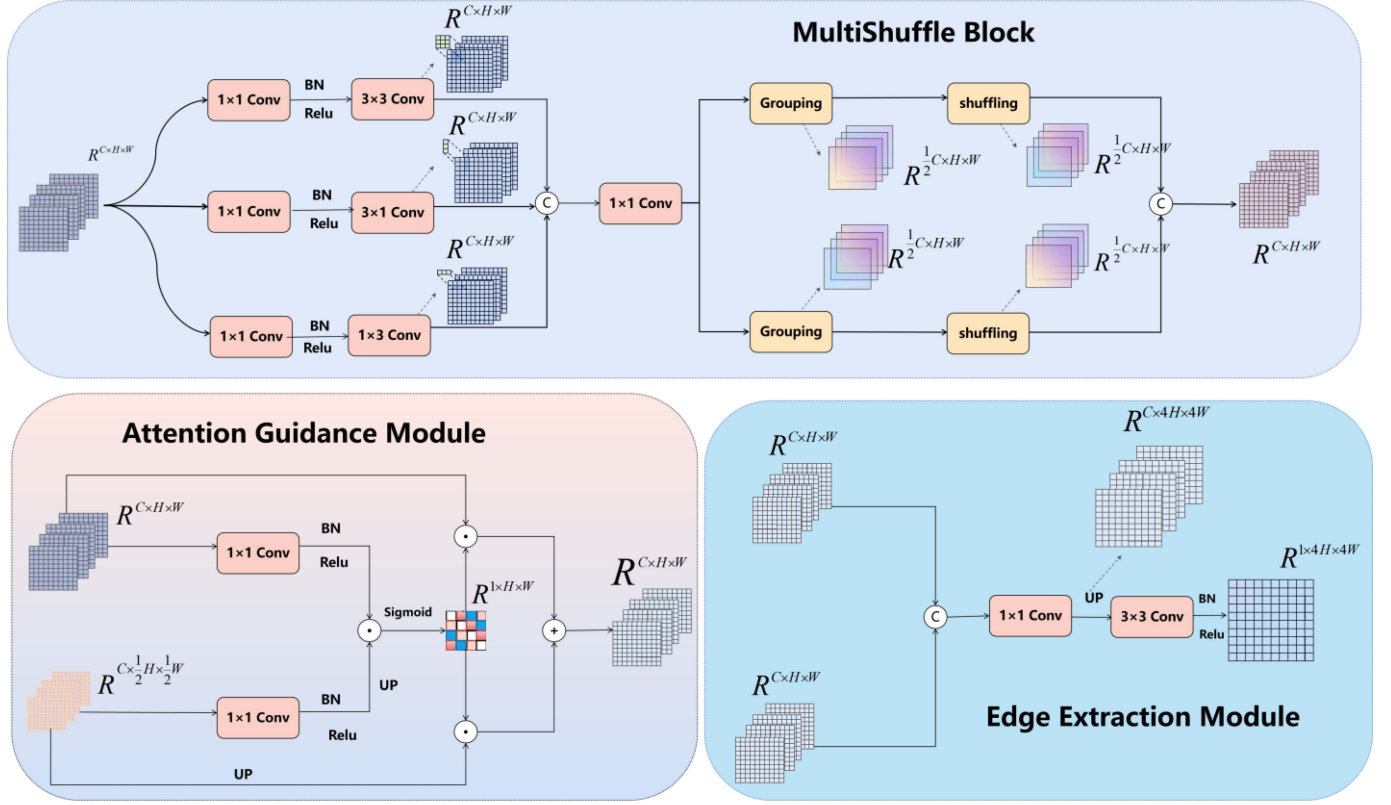


Fig. 3. Details of the MultiShuffle block. Details of the attention guidance module and edge extraction module.

rectify structural inconsistencies in homogeneous areas, while detail features refine edge fidelity in heterogeneous zones.

Architecturally, the DSSDB undergoes resolution adaptation before AGM integration: pre-AGM convolutional layers employ stride-2 operations for dimension reduction, transitioning to stride-1 convolutions post-AGM to maximize information retention. The output of AGM is formulated as

$$O_{u \text{ t}_{\text{Agm}}} = \sigma S F + (1 - \sigma) D F. \quad (2)$$

Studies suggest that the features in the detail branch [45], [46], [47], which have a higher resolution, can better preserve spatial details. The EEM mitigates false edge artifacts in complex scenes by hierarchically fusing multiscale detail features. Through concatenation, bilinear upsampling, and 1×1 convolutions, EEM (see Fig. 3) generates edge features EF enriched with global context, ensuring accurate boundary localization even in heterogeneous regions.

C. Spatial-Channel Dominated Semantic Branch

The SCDSB operates in parallel with the DSSDB to hierarchically aggregate global contextual semantics. To efficiently capture long-range dependencies and multiscale patterns crucial for complex remote sensing scenes (e.g., handling extreme scale variations from sub-meter vehicles to kilometer-scale fields), we introduce the LAPPM as the core component of SCDSB. LAPPM synergizes the strengths of global interaction modeling

and multiscale context aggregation through two key, codesigned mechanisms.

- 1) *Cascaded Linear Attention*: Replaces quadratic-complexity dot-product operations with first-order Taylor approximation, enabling efficient long-range dependency modeling across spatial and channel dimensions.
- 2) *Multibranch Kernel-Sharing Convolution*: Employs parallel depthwise dilated convolutions (rates = 12,3) with complementary 3×3 and 5×5 kernels, capturing multiscale patterns while maintaining computational efficiency through parameter sharing.

The cascaded linear attention mechanism in LAPPM orchestrates bidirectional spatial-channel context integration through two synergistic phases. Initially, cross-dimensional dependency modeling establishes long-range correlations between channel-wise statistics and spatial positions, generating channel-priority semantic priors. These priors subsequently guide position-aware feature recalibration, where spatial semantics are refined through attention-weighted positional relationships.

Concretely, the high-resolution feature map preserves fine-grained details while two distinct low-resolution counterparts undergo bilinear upsampling and element-wise summation to form enriched low-resolution features. Both of them are then processed through coordinate attention [39] to jointly optimize channel sensitivity and positional awareness.

The refined features are projected into query (Q), key (K), and value (V) matrices. Diverging from quadratic-cost dot-product attention [48], we implement the first-order Taylor

approximation

$$e^{q_i^T k_j} \approx 1 + q_i^T k_j. \quad (3)$$

To ensure that the above-mentioned approximation is non-negative, we also apply L2 normalization. The i th row of result matrix generated by dot-product attention can be written as (4), and the introduced linear attention can be written as (5)

$$D(Q, K, V)_i = \frac{\sum_{j=1}^N e^{q_i^T k_j} v_j}{\sum_{j=1}^N e^{q_i^T k_j}} \quad (4)$$

$$D(Q, K, V)_i = \frac{\sum_{j=1}^N v_j + \left(\frac{q_i}{\|q_i\|_2}\right)^T \sum_{j=1}^N \left(\frac{k_j}{\|k_j\|_2}\right) v_j^T}{N + \left(\frac{q_i}{\|q_i\|_2}\right)^T \sum_{j=1}^N \left(\frac{k_j}{\|k_j\|_2}\right)} \quad (5)$$

where i represents the rows of the vector and j represents the columns of the vector.

Equation (5) can be transformed into a vectorized form

$$D(Q, K, V) = \frac{\sum_j V_{i,j} + \left(\frac{Q}{\|Q\|_2}\right)^T \left(\left(\frac{K}{\|K\|_2}\right)^T V\right)}{N + \left(\frac{Q}{\|Q\|_2}\right)^T \sum_j \left(\frac{K}{\|K\|_2}\right)_{i,j}}. \quad (6)$$

Since $\sum_{j=1}^N \left(\frac{k_j}{\|k_j\|_2}\right) v_j^T$ and $\sum_{j=1}^N \left(\frac{k_j}{\|k_j\|_2}\right)$ can be computed and reused for each query, the time and memory complexity based on the aforementioned approach is linear.

In the latter part of LAPPM, we utilize the multibranch Kernel-sharing convolution to further expand the receptive field and capture multiscale semantic information. The three branches employ depthwise dilated convolutions [49] with dilation rates of 1, 2, and 3, respectively. However, during the downsampling process using dilated convolutions, each newly obtained pixel is composed of nonadjacent pixels, which affects the continuity of information flow. In addition, the gaps within the convolution kernel often lead the network to neglect small targets. To alleviate the aforementioned issue, we applied an additional convolution operation to the last two branches, using depthwise convolutions with kernel sizes of 3 and 5, respectively. Finally, we concatenated the feature maps from the three branches to merge multiscale information, followed by a 1×1 convolution to adjust the channel to the input size. Notably, except for the final convolution, all other convolution operations do not alter the channel count or resolution.

D. Edge-Guided Feature Fusion Decoder

The divergent feature extraction strategies employed by the DSSDB and SCDSB inherently lead to representational discrepancies: detail features retain high-resolution spatial precision but lack semantic coherence, while SFs encapsulate global context at the cost of localized granularity. To effectively harmonize these complementary representations, we propose the FFM (see Fig. 4). Initially, edge features generated by the DSSDB's EEM compute spatially variant attention weights through 1×1 convolution and sigmoid activation

$$\text{edgeatt} = \text{sigmoid}(\text{Maxpool}(\text{conv}_{1 \times 1}(\text{EF}))). \quad (7)$$

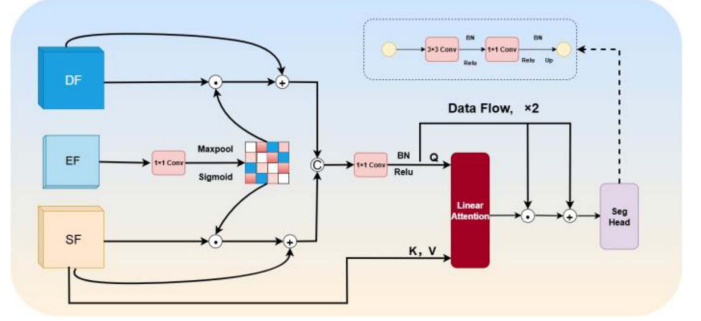


Fig. 4. Details of FFM, where “Maxpool” indicates that the feature map has undergone max pooling for downsampling, “DF” represents detailed features from DSSDB, “SF” represents semantic features from SCDSB, and “EF” represents edge features.

The edge attention is used to apply weights to the DF and SF separately. Due to the guiding effect of the edge information, the disparity between DF and SF is reduced. After passing through a residual structure, the two feature maps are concatenated to obtain a fused feature (FF). This process can be expressed by the following equation:

$$FF = \text{Conv}_{1 \times 1}$$

$$\text{Concat}(\text{DF} \times \text{edgeatt} + \text{DF}, \text{SF} \times \text{edgeatt} + \text{SF}). \quad (8)$$

Next, we use the FF map as the query (Q), while the SF map serves as the key (K) and value (V). Through linear attention, the FF guides SF on which information to focus on, and this information is effectively integrated into the FF map. Finally, a segmentation head, consisting of two convolutional layers and one upsampling layer, is used to adjust the image resolution to match the input image size and the channels to the number of pixel classes, producing the final segmentation result.

E. Loss Function

For the proposed method, the total loss function consists of two parts: the main loss and the edge prediction loss. Due to the class imbalance in the land segmentation dataset, we combine cross-entropy loss and Dice loss to form the main loss. The equations are as follows:

$$\mathcal{L}_{CE} = - \sum_{c=1}^C y_c \log(p_c) \quad (9)$$

$$\mathcal{L}_{D[.]]= 1 - \frac{2 \sum_{i=1}^N p_i y_i}{\sum_{i=1}^N p_i + \sum_{i=1}^N y_i} \quad (10)$$

$$\mathcal{L}_{main} = \mathcal{L}_{CE} + \mathcal{L}_{D[.]]= \quad (11)$$

For the edge loss, we employ the traditional binary cross-entropy loss, treating edges and nonedges as two distinct classes. The total loss function can be expressed by the following equation:

$$\mathcal{L}_{edge} = - \frac{1}{N} \sum_{i=1}^N [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)], \quad (12)$$

$$\mathcal{L} = w_1 \mathcal{L}_{\text{main}} + w_2 \mathcal{L}_{\text{edge}}. \quad (13)$$

Since the main loss and the edge loss differ in terms of the number of classes and segmentation difficulty, we assign two weight parameters to balance their impact. Through experimental validation, we set $w_1 = 0.7$ and $w_2 = 0.3$ as the final values.

IV. DATASET AND EXPERIMENTAL SETTING

A. Datasets

The DeepGlobe land cover dataset [50] comprises RGB satellite imagery with an original resolution of 2448×2448 pixels, accompanied by corresponding semantic masks for land cover annotation. This dataset encompasses six land cover categories: “Urban,” “Farmland,” “Mountain,” “Forest,” “Water,” and “Barren.” The inherent complexity arising from diverse terrain features and fine-grained class distinctions presents greater challenges compared to the ISPRS Wageningen and Potsdam datasets. To expand the original dataset of 803 images, we applied a cropping approach that generated 3212 samples at 1224×1224 pixels resolution through center-cropped preprocessing. These data were randomly partitioned into training, validation, and test sets following a 6:2:2 ratio.

The LoveDA Dataset [51] contains 5987 high-resolution (1024×1024 pixels) satellite images captured across three Chinese urban centers: Nanjing, Changzhou, and Wuhan. Distinguished by its substantial pixel-level samples and comprehensive object annotations, this dataset features seven land cover classes: “Building,” “Road,” “Lake,” “Bare Land,” “Forest,” “Farmland,” and “Background.” Given the unavailability of test set annotations, we consolidated the original 2522 training images and 1669 validation images into a unified dataset. This combined collection was subsequently reorganized through random sampling into new training, validation, and test subsets using a 4:3:3 distribution ratio, ensuring equitable representation across all land cover categories.

It is worth mentioning that to enable edge-aware supervision without explicit boundary annotations, we generate pseudo-edge labels directly from ground truth segmentation masks. This is achieved through morphological processing using the Canny edge detector with empirically determined thresholds.

B. Implementation Details

To accommodate large network architectures within GPU memory constraints, we standardized input dimensions by resizing all images to 512×512 pixels, supplemented by data augmentation techniques including random flipping, Gaussian blur filtering, and color space transformations. All experiments employ stochastic gradient descent optimization configured with momentum (0.9), weight decay (0.0001), and an initial learning rate of 0.007. The training protocol comprises 500 epochs per dataset with a consistent batch size of 10 samples, ensuring comparative stability across experimental conditions. The computational platform features an Intel Core i7-10700K octa-core processor (2.9 GHz) coupled with an Nvidia GeForce GTX 4060Ti GPU (16 GB VRAM). The software environment is implemented within PyTorch 2.21 framework accelerated by

CUDA 12.1, leveraging parallel computing capabilities for efficient model training.

C. Evaluation Metrics

We use the generalized segmentation metrics of semantic segmentation algorithms to evaluate the network’s performance, including mean accuracy (mAcc), mean intersection over union (mIoU), class-wise intersection over union (IoU), mean F1 score (mF1), number of parameters (Params), and frames per second (FPS). Some of the calculation formulas are as follows:

$$\text{mAcc} = \frac{1}{N} \sum_{i=1}^N \frac{\text{TP}_i + \text{TN}_i}{\text{TP}_i + \text{FN}_i + \text{TN}_i + \text{FP}_i} \quad (14)$$

$$\text{mIoU} = \frac{1}{N} \sum_{i=1}^N \frac{\text{TP}_i}{\text{TP}_i + \text{FP}_i + \text{FN}_i} \quad (15)$$

$$\text{F1} = 2 \times \frac{P \times R}{P + R} \quad (16)$$

$$P = \frac{\text{TP}}{\text{TP} + \text{FP}}. \quad (17)$$

Here, N represents the total number of categories in the dataset, and i denotes the index of each category. TP_i represents the number of pixels of category i correctly classified, FP_i represents the number of pixels of category i misclassified as other categories in the predicted map, FN_i denotes the number of pixels from other categories misclassified as category i and TN_i indicates the number of pixels from categories other than i correctly classified.

V. EXPERIMENTAL RESULTS

To evaluate the efficacy of the proposed methodology, we conduct a comprehensive comparative analysis against state-of-the-art semantic segmentation frameworks on both the DeepGlobe Land Cover [50] and LoveDA [51] datasets. The benchmarked methods encompass: DeepLabv3+ [13], a classical CNN-based semantic segmentation method; ABCNet [26], utilizing dual-branch structural design; LCNet [52], integrating three-branch contextual aggregation; PidNet [28] with explicit edge-aware learning; Unetformer [53], a transformer-based semantic segmentation approach; Paraformer [54], using low-resolution land cover data to guide high-resolution land cover mapping. And DeepKANSeg [55], including two key innovations based on the emerging Kolmogorov–Arnold Network (KAN). Quantitative evaluations on the DeepGlobe dataset (see Table I) and LoveDA dataset (see Table II) demonstrate systematic performance comparisons across multiple metrics.

A. Experimental Results on the DeepGlobe Land Cover Dataset

Table I shows the experimental results on the DeepGlobe Land Cover dataset [50]. The proposed method achieves the best performance with 88.14% mAcc, 73.9% mIoU, and 82.65% recall, while maintaining model efficiency with 13.02 million

TABLE I
QUANTITATIVE COMPARISON OF RESULTS ON THE DEEPGLOBE LAND COVER DATASET WITH COMPETING METHODS

Method	mAcc(%)	mIoU(%)	Recall(%)	Params(M)	FPS (1224*1224)
Deeplabv3+[13]	86.64	69.91	80.39	40.35	13.34
ABCNet [26]	86.85	69.04	78.60	13.39	15.62
Unetformer [53]	87.34	70.95	80.77	13.68	16.35
PidNet [28]	87.61	70.90	81.38	28.54	33.23
LCNet [52]	83.42	63.53	73.91	0.71	19.76
Paraformer [54]	86.67	70.35	81.22	108.57	7.02
DeepKANSeg [55]	86.99	71.81	81.49	43.50	10.34
ESCNet	88.14	73.90	82.65	13.02	21.37

The bold entities indicate the best results.

TABLE II
QUANTITATIVE COMPARISON OF RESULTS ON THE LOVEDA DATASET WITH COMPETING METHODS

Method	mAcc(%)	mIoU(%)	Recall(%)	Params(M)	FPS(1024*1024)
Deeplabv3+[13]	71.39	56.93	72.13	40.35	20.93
ABCNet [26]	72.57	58.46	71.93	13.39	28.23
Unetformer [53]	72.40	58.21	71.11	13.68	27.58
PidNet [28]	71.76	56.87	70.43	28.54	53.60
LCNet [52]	71.72	57.69	71.78	0.71	27.25
Paraformer [54]	71.50	57.62	71.87	108.57	9.39
DeepKANSeg [55]	71.14	57.85	72.75	43.50	15.77
ESCNet	72.98	59.32	72.80	13.02	37.51

The bold entities indicate the best results.

TABLE III
CATEGORY-WISE DISTRIBUTION AND QUANTITATIVE RESULTS OF SEGMENTATION ON THE DEEPGLOBE LAND COVER DATASET

Method	Category IoU(%)						mIoU	mF1
	Urban	Agriculture	Range	Forest	Water	Barren		
Proportion (%)	9.35	56.76	10.21	13.75	3.74	6.14		
Deeplabv3+[13]	75.14	85.58	36.19	78.8	75.30	68.46	69.91	83.40
ABCNet [26]	75.00	86.03	31.56	79.47	73.01	69.17	69.04	82.52
Unetformer [53]	75.49	86.07	40.00	81.32	75.08	67.75	70.95	83.93
PidNet [28]	75.91	86.33	38.22	82.02	76.50	66.43	70.90	84.38
LCNet [52]	72.46	82.89	25.62	69.64	71.23	59.38	63.53	78.38
Paraformer [54]	77.12	87.83	38.24	77.29	79.51	66.12	71.02	83.86
DeepKANSeg [55]	74.98	87.31	40.51	79.79	77.92	70.37	71.81	84.15
ESCNet	77.29	88.42	44.00	82.24	79.72	71.72	73.90	85.31

The bold entities indicate the best results.

parameters. In terms of inference speed, the framework processes 21.37 FPS for 1224×1224 resolution images.

In the DeepGlobe land cover dataset, we compare our method with others for each individual category, as shown in Table III. The results indicate that, compared to other methods, the proposed approach performs better at recognizing small objects, such as rivers and urban buildings. This advantage can be attributed to several factors. First, we refine detailed information by incorporating boundary information, which helps the model better capture the boundaries between different objects and

subtle differences, reducing boundary blurring. In addition, the introduction of the LAPP in the semantic branch enables the model to capture multiscale information, which further benefits the segmentation of small objects.

To comprehensively evaluate the proposed method against existing methods in terms of segmentation performance, we selected some representative images (see Fig. 5). In these images, vegetation grows around the water bodies, and the color of the river water is relatively light, which easily leads to confusion between water bodies and pasture categories. Therefore,

TABLE IV
CATEGORY-WISE DISTRIBUTION AND QUANTITATIVE RESULTS OF SEGMENTATION ON LOVE-DA DATASET

Method	Category IoU(%)							mIoU	mF1
	Background	Building	Road	Water	Barren	Forest	Agriculture		
Proportion (%)	34.72	9.20	4.86	8.24	4.71	12.17	22.78		
Deeplabv3+[13]	53.65	53.07	53.9	69.69	39.96	60.91	67.31	56.93	71.76
ABCNet [26]	56.87	54.84	54.65	70.72	40.46	62.76	68.90	58.46	72.25
Unetformer [53]	57.57	50.51	56.52	70.79	41.92	61.80	68.33	58.21	71.75
PidNet [28]	56.38	51.12	54.58	70.76	37.66	60.94	66.69	56.87	71.09
LCNet [52]	55.87	54.50	55.52	71.42	37.95	61.18	67.42	57.69	71.75
Paraformer [54]	55.59	53.54	53.10	70.41	40.44	62.50	67.77	57.62	71.68
DeepKANSeg [55]	57.14	54.33	53.24	70.93	39.62	62.69	67.08	57.85	71.93
ESCNet	57.15	56.39	57.40	71.60	40.99	63.18	68.52	59.32	72.89

The bold entities indicate the best results.

successfully identifying the outline of water bodies is a challenging task. In Image-1, only our ESCNet and MKANet have roughly restored the outline of the water bodies. LCNet and Paraformer struggled to identify the outline of the river, while ABCNet, PidNet, and Unetformer incorrectly classified the river as pasture. In addition, due to the presence of a road in the middle of the river, splitting the river into upper and lower parts, MKANet and Deeplabv3+ failed to accurately restore the lower part of the river and mistakenly classified parts of the river as pasture. Our ESCNet effectively segmented the images, successfully identifying the river that was interrupted by the road.

B. Experimental Results on the LoveDA Dataset

To evaluate the performance of the proposed model, we conducted further experiments on the LoveDA [51] dataset. The experimental results once again validate the superiority of our method, achieving the best segmentation accuracy. Specifically, the method reaches 72.98% in mAcc, 59.32% in mIoU, and 72.80% in recall. In addition, we measured the inference speed on images from the LoveDA dataset, where the inference speed is 37.51 FPS for the 1024×1024 resolution images.

Similarly, we performed a per-category comparison on the LoveDA dataset, as shown in Table IV. The experimental results show that while achieving comparable performance on large objects (agriculture: 68.52% vs. 68.9% in ABCNet [26]), ESCNet demonstrates superior small-object segmentation (buildings: 56.39% vs. 54.50% in LCNet [52]),

Fig. 6 shows the visualization results of the dataset, where it can be observed that our method has more completely restored the outlines of water bodies and buildings within the selected areas, reducing the phenomenon of object confusion caused by complex backgrounds.

C. Ablation Study

In this section, we conduct an ablation study on ESCNet to evaluate the effectiveness of the proposed components. The

TABLE V
QUANTITATIVE RESULTS OF THE ABLATION RESEARCH

BaseLine	SCDSB	DSSDB	EGFFD	FPS	mIoU(%)
√				29.26	70.34
√	√			22.44	72.06
√	√	√		22.09	72.53
√	√	√	√	21.37	73.9

The bold entities indicate the best results.

comparative experiments were performed on the DeepGlobe land cover dataset, using the same environment and inference settings. The baseline model removes all the proposed modules and replaces them with alternative methods. For DSSDB, we replace the MultiShuffle Block and AGM with 3×3 convolutions and concatenation operations. For SCDSB, the LAPP module—an integrated plug-and-play component—was removed entirely. Finally, regarding EGFFD, we employ addition operations instead to achieve branch fusion. Comprehensive details of the ablation studies and quantitative results are presented in Table V, with corresponding qualitative prediction results shown in Fig. 7.

Baseline: We use ResNet-18 as the encoder for the SCDSB, remove the LAPP, and the edge information extracted in the DSSDB is only used as a supplement to the detail information, without participating in the fusion of the two branches. The feature maps generated by the two branches are fused using concatenation.

Baseline + SCDSB: To establish long-range dependencies while balancing computational cost and parameter count, we propose SCDSB. LAPP is deployed in the SCDSB and, with a slight increase in the number of parameters, achieves a significant improvement in mIoU, with a 1.72% increase. The experimental results demonstrate that SCDSB effectively establishes long-range dependencies.

Baseline + SCDSB + DSSDB: The DSSDB is designed to capture multidirectional spatial details, from which the edge features are further derived through our EEM. As demonstrated in

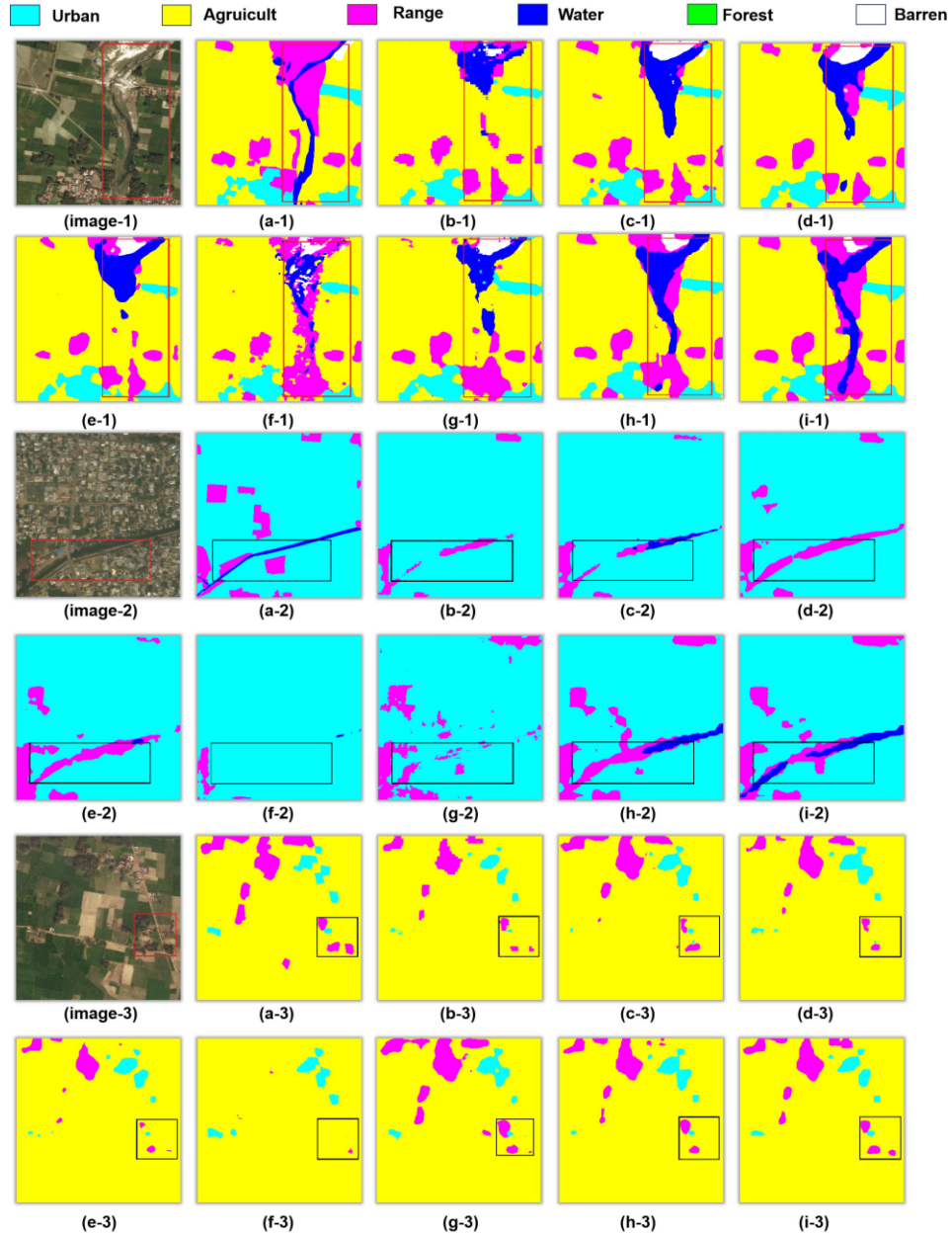


Fig. 5. Qualitative prediction results of various methods on the DeepGlobe land cover dataset. (a) Ground truth. (b) ABCNet [26]. (c) Deeplabv3+ [13]. (d) PidNet [28]. (e) Unetformer [53]. (f) LCNet [52]. (g) Paraformer [54]. (h) DeepKANSeg [55]. (i) ESCNet. The following numbers represent the corresponding picture.

Table V, integrating DSSDB into the Baseline + SCDSB framework achieves a 0.47% improvement in mIoU while maintaining computational efficiency, with only negligible FPS degradation. This validates the effectiveness of our approach in preserving critical spatial information for edge-aware feature extraction without compromising inference speed.

Baseline + SCDSB + DSSDB + EGFFD: To validate the cross-branch fusion capability of EGFFD, we integrate it into the Baseline + SCDSB + DSSDB model. Compared to the previous configuration, mIoU further improves by 1.37%. These results indicate that the proposed modules work synergistically, where EGFFD effectively mitigates the discrepancies between low-level and high-level feature representations through edge-aware guidance.

VI. DISCUSSION

In the corresponding experiments, ESCNet attained the highest segmentation accuracy on the DeepGlobe land cover and LoveDA datasets. We further tested the model's parameters and inference speed. The comparative data in tables indicate that ESCNet achieves better segmentation accuracy with a competitive parameter count. This efficiency stems from our optimization strategies during contextual semantic extraction, such as linear attention and shared convolutional kernels, which reduce computational complexity while maintaining performance. While not the fastest in absolute inference speed, ESCNet strikes an effective balance between accuracy and efficiency, making it suitable for practical scenarios where both metrics are prioritized.

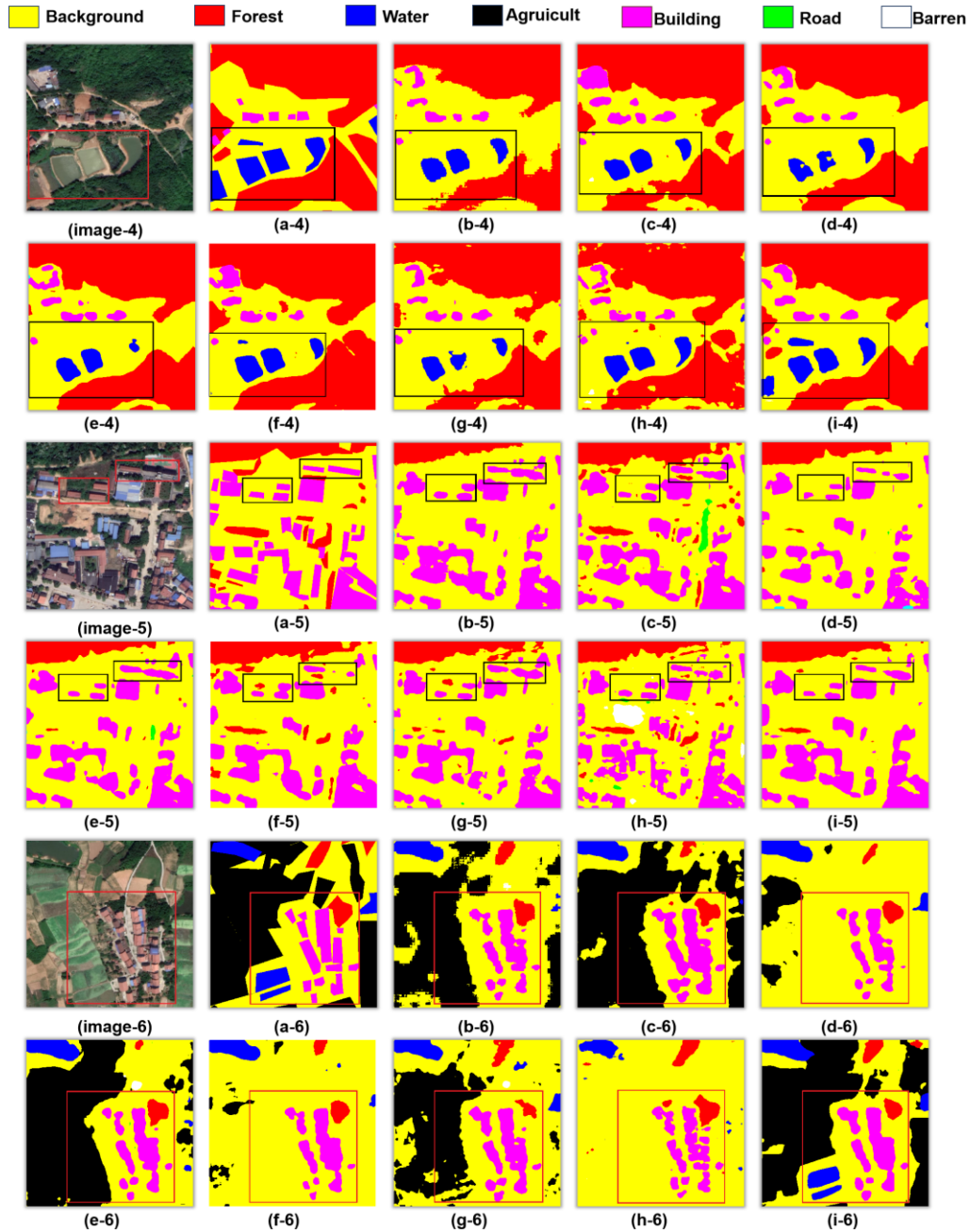


Fig. 6. Qualitative prediction results of various methods on the LoveDA dataset. (a) Ground truth. (b) ABCNet [26]. (c) Deeplabv3+ [13]. (d) PidNet [28]. (e) Unetformer [53]. (f) LCNet [52]. (g) Paraformer [54]. (h) DeepKANSeg [55]. (i) ESCNet. The following numbers represent the corresponding picture.

ESCNet's ability to capture precise edge information is another critical factor contributing to its high segmentation accuracy. Edge details are treated as complementary to fine-grained features and assist in dual-branch information fusion. DSSDB acquires multidirectional spatial details through asymmetric convolution. For extracting edge information, this process also alleviates the interference of internal pixels of the object to a certain extent.

Furthermore, our hierarchical feature fusion strategy plays a crucial role in achieving robust segmentation performance. As shown in Fig. 8, the corresponding heatmaps visualize the

network's focus regions. It can be observed that the EEM primarily focuses on global contours rather than internal categorical details of targets. The SCDSB specializes in capturing small, fragmented objects; however, it exhibits significant inconsistency in attending to intraclass instances. In contrast, the DSSDB emphasizes large targets but frequently neglects small ones. Therefore, by leveraging the edge features extracted from the EEM as guidance to integrate these complementary characteristics, our FFM module ultimately achieves refined segmentation results—precisely capturing small targets while simultaneously maintaining clear boundary contours.

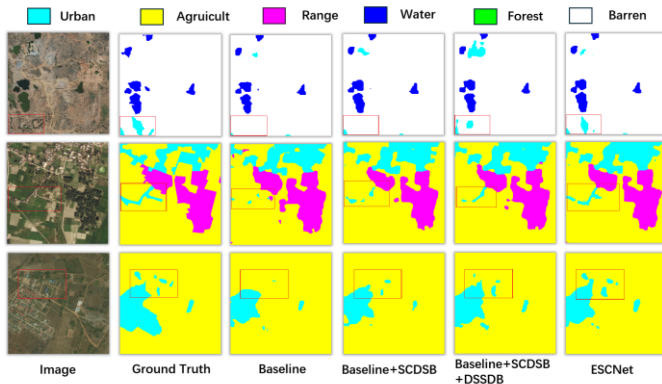


Fig. 7. Qualitative results of the ablation study.

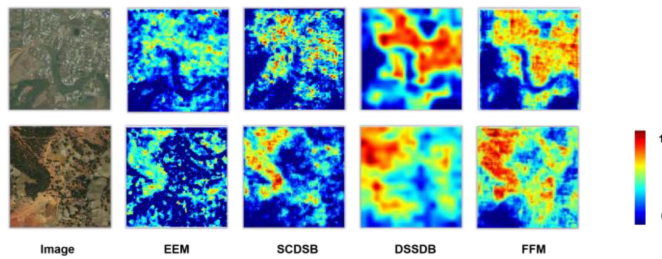


Fig. 8. Comparison of modality weights generated by the penultimate layer of EEM, the last layer of SCDSB, the last layer of DSSDB, and FFM.

Despite the advantages discussed above, a potential limitation arises from our backbone selection within the SCDSB. To maintain a competitive parameter count, we employed ResNet as the backbone rather than exploring architectures specifically designed for long-range dependency modeling, such as Swin Transformers or Mamba. Although effective in our experiments, this design choice introduces inherent constraints on the model's ability to capture highly complex global contextual relationships compared to these newer architectures. This represents a trade-off between parameter efficiency and the richness of extracted high-level semantics.

VII. CONCLUSION

To address the critical challenge of harmonizing high-level semantics with spatial details in land cover classification, this study proposes ESCNet. The DSSDB preserves fine-grained textures and directional patterns through asymmetric convolutions and channel shuffling, while the SCDSB enables spatial-channel interactive reasoning, capturing discriminative hierarchical semantics. The EGFFD dynamically aligns these complementary representations using edge-attention mechanisms, effectively mitigating boundary ambiguities and intraclass inconsistencies prevalent in conventional fusion approaches. The experimental results demonstrate that ESCNet provides a viable approach for integrating edge guidance in dual-branch segmentation architectures. This methodology shows potential applicability for improving land cover classification accuracy, particularly in scenarios requiring precise boundary delineation. In the future, we aim to apply ESCNet to more complex scenarios. In addition, with the increasing number of multimodal remote sensing data,

we will explore the possibility of semantic segmentation based on cross-modal data.

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